



Presentation of the Deep Learning Project

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Contents

1. Introduction
2. Data Preparation and Preprocessing
3. Exploratory Data Analysis
4. Basic Machine Learning Models
5. Graph Convolutional Networks (GCN) Model
6. Conclusions



Part 1: Introduction

Key objectives:

- Introduce the Problem and the Dataset.



Dataset

This dataset offers a comprehensive collection of attributes for various animal species worldwide, featuring 205 samples before preprocessing.

Tabel 1: Dataset Samples															
Animal	Height (cm)	Weight (kg)	Color	Lifespan (years)	Diet	Habitat	Predators	Average Speed (km/h)	Countries Found	Conservation Status	Family	Gestation Period (days)	Top Speed (km/h)	Social Structure	Offspring per Birth
Aardvark	105-130	40-65	Grey	20-30	Insectivore	Savannas, Grasslands	Lions, Hyenas	40	Africa	Least Concern	Orycteropodidae	210-240	40	Solitary	1
Aardwolf	40-50	14-Aug	Yellow-brown	12-Oct	Insectivore	Grasslands, Savannas	Lions, Leopards	24-30	Eastern and Southern Africa	Least Concern	Hyaenidae	90	40	Solitary	5-Feb



Problem

The primary task associated with this dataset involves data cleaning and classification of animals into three categories based on their diet: Omnivore, Carnivore, and Herbivore.

CARNIVOROUS		HERBIVOROUS		OMNIVOROUS	
					
Tiger	Cuttlefish	Rhinoceros	Deer	Hedgehog	Boar
					
Platypus	Anteater	Parrot	Koala	Skunk	Monkey
					
Shark	Lion	Hippopotamus	Giraffe	Kiwi	Bear



Part 2: Data Preparation and Preprocessing

Key objectives:

- Handle missing values
- Address incorrect data
- Encode categorical variables
- Normalize/Standardize the dataset



Missing Values

- Approximately 20 percent of the data contains at least one missing value, making it crucial to handle these gaps appropriately.
- Deleting these rows was not an option, as it would significantly reduce the dataset size.
- Given the characteristics of the dataset, we can impute the missing values using ChatGPT.
- After preprocessing, we have 187 non-null data samples.



Transforming Targets

We grouped similar categories in the target label based on domain knowledge, reducing the total from 16 to 3 main categories.

Tabel 2: Targets	
Diet	Count
Carnivore	94
Herbivore	51
Omnivore	42



Final Numeric Data Description

	Height (cm)	Weight (kg)	Lifespan (years)	Average Speed (km/h)	Gestation Period (days)	Offspring per Birth
count	187.000000	187.000000	187.000000	187.000000	187.000000	187.000000
mean	175.828877	1889.816746	22.933983	31.377995	134.829259	1182.791444
std	373.586782	15409.975050	18.961394	22.355671	129.028395	14636.520588
min	4.500000	0.006000	0.071429	0.025000	0.071429	1.000000
25%	40.000000	3.325000	11.500000	14.000000	37.000000	1.000000
50%	78.500000	31.500000	17.500000	30.000000	90.000000	2.500000
75%	150.000000	180.000000	27.500000	45.750000	207.500000	15.750000
max	3200.000000	200000.000000	125.000000	120.000000	650.000000	200000.000000



Part 3: Exploratory Data Analysis

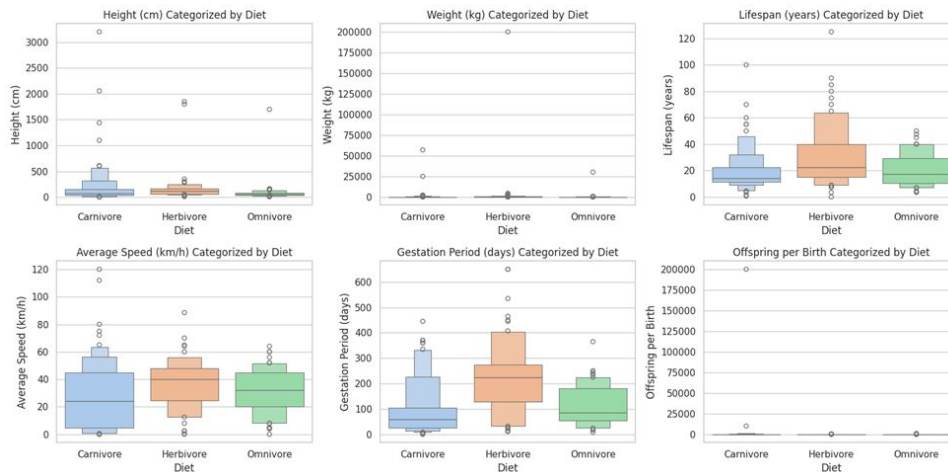
Key objectives:

- Data Visualization
- Identifying Patterns and Relationships
- Detecting Anomalies and Outliers
- Relationships and Correlations



Visualization of numerical data features (1)

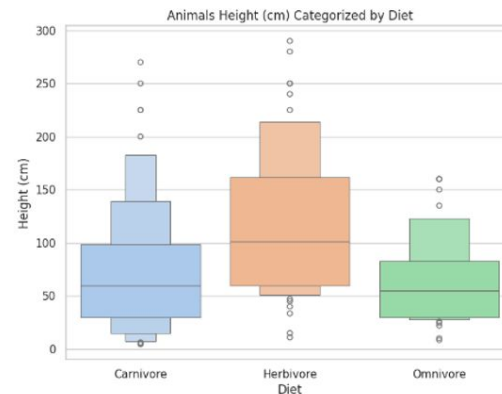
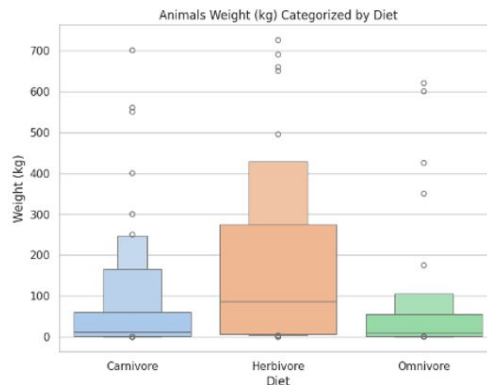
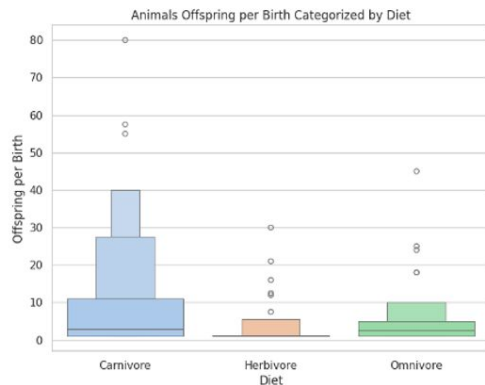
As you can see, outlier data affects the quality of the charts and can also impact the accuracy of classification.





Visualization of numerical data features (2)

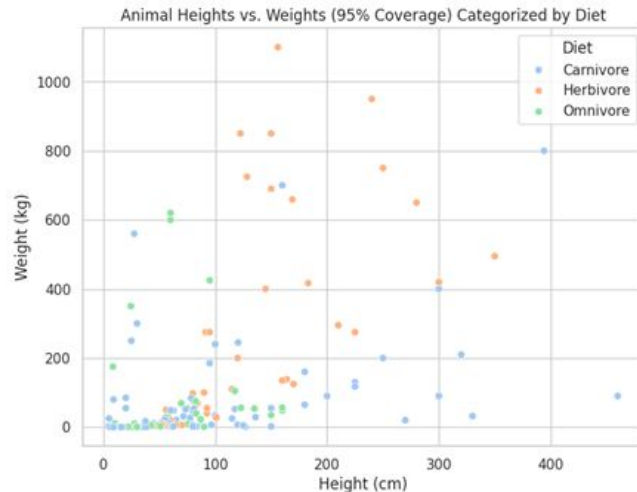
We can now view the precise boxplot, which includes 95 percent data coverage.





Visualization of numerical data features (3)

This chart illustrates the correlation between animal size and their diet, which includes 95 percent data coverage.

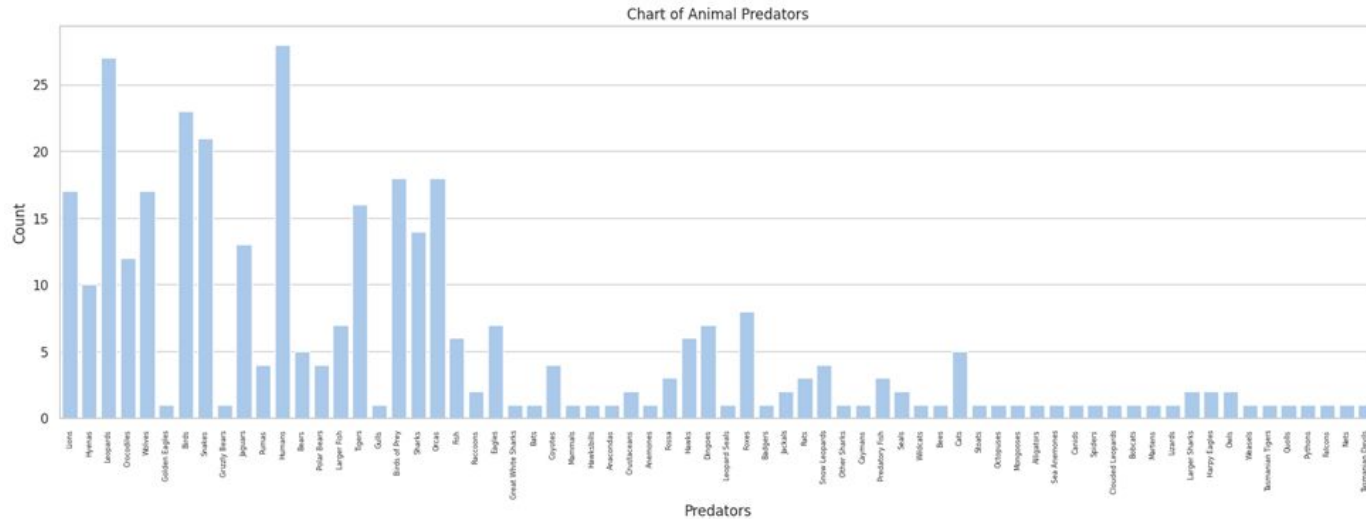






Visualization of animal's predators

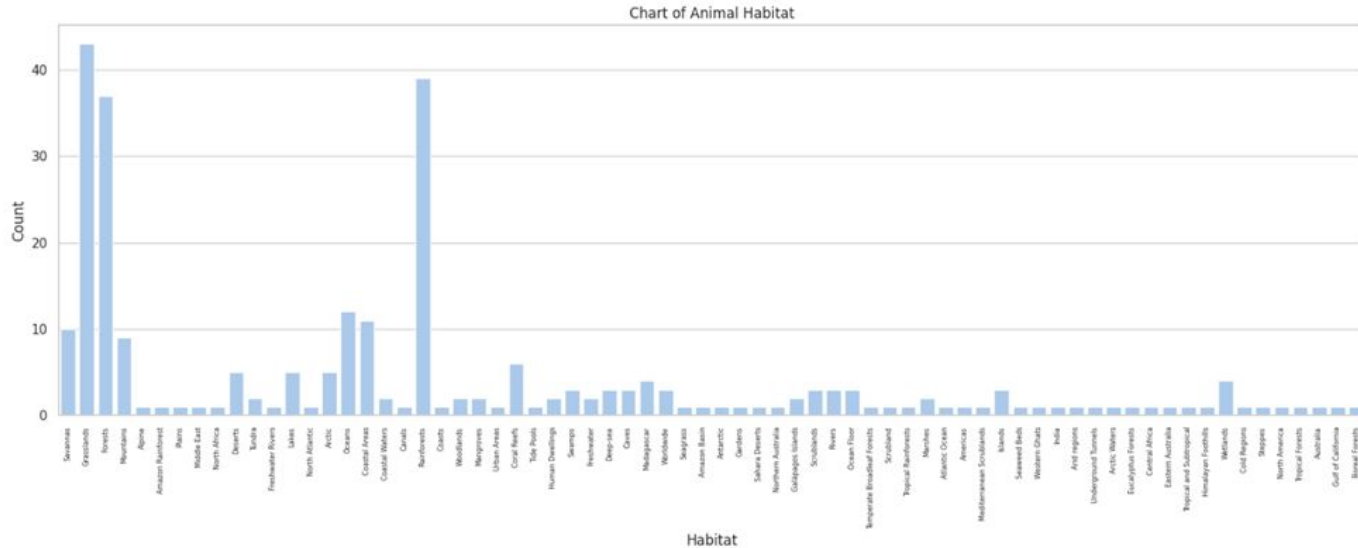
An interesting fact is that humans are the top predators of most animal species.





Visualization of animal's predators

Most of the animals in the dataset inhabit forests.





Part 4: Basic Machine Learning Modes

Key objectives:

- Implement five fundamental machine learning models
- Report the accuracy of each model on the same test set



Machine Learning Models

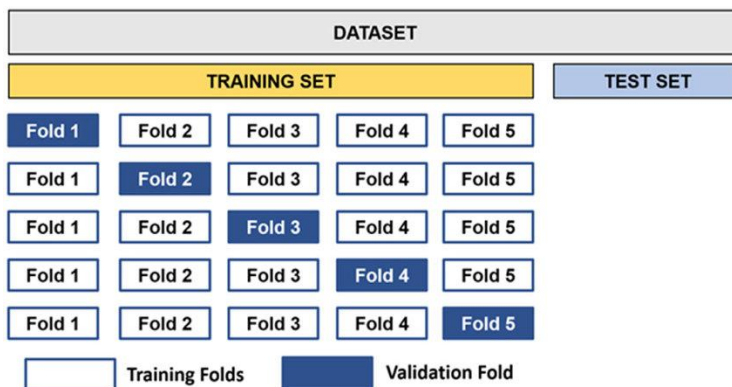
We evaluate and compare the accuracy of five machine learning models on a dataset. The models used are:

1. Support Vector Machine (SVM)
2. Random Forest (RF)
3. Logistic Regression (LR)
4. Extreme Gradient Boosting (XGBoost)
5. Multilayer Perceptron (MLP)



Validation

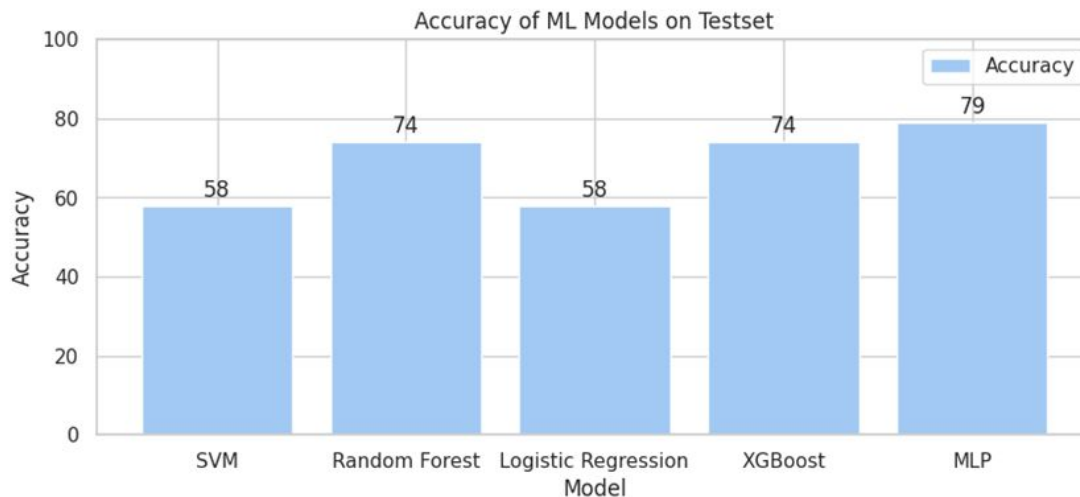
To validate our models, we allocate 10 percent of the data for testing, a practice consistent across all models. We then employ 9-fold cross-validation, using 10 percent of the data for validation and 90 percent for training in each fold. This approach helps identify potential overfitting due to the small size of the dataset.





Accuracy on Test dataset

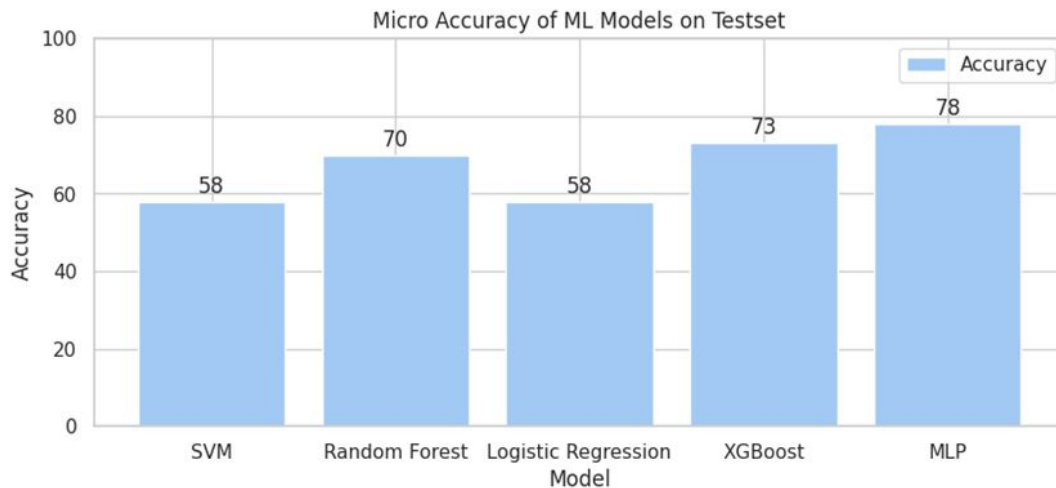
"As you can see, the MLP model performs better in classifying the test dataset.





Macro Accuracy on Test dataset

Given the imbalance in the dataset, macro accuracy is a better metric for evaluating classifier performance. As observed, the accuracy of the random forest model decreases using this method.





Part 5: Graph Convolutional Networks (GCN) Model

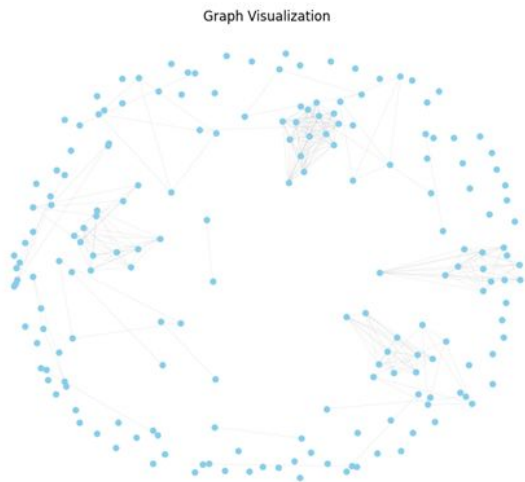
Key objectives:

- Design a graph using the Tabular dataset
- Implement the GCN model
- Evaluate the model and compare its results with previous models



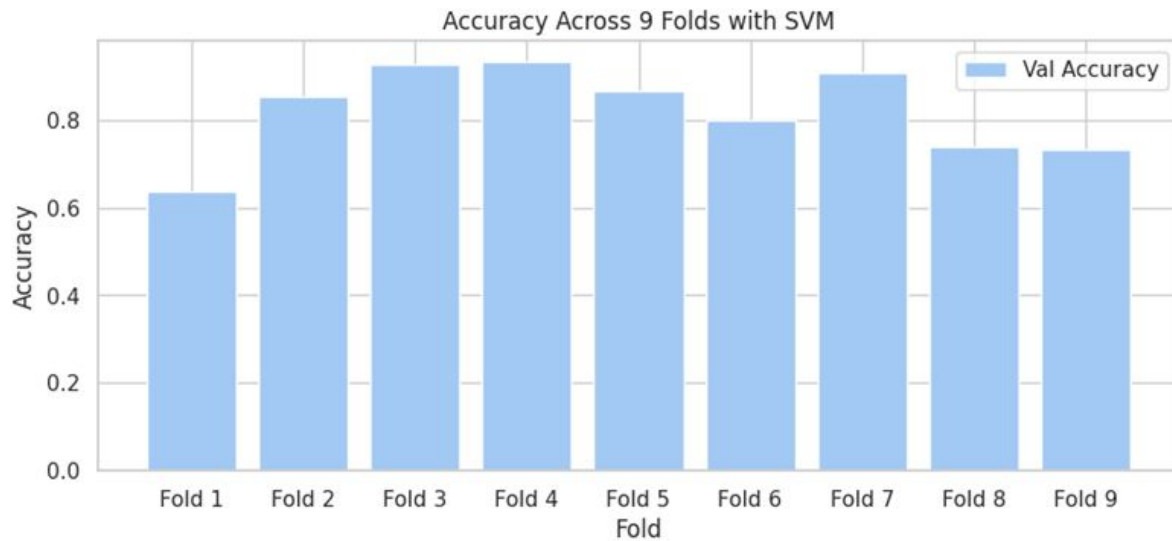
Graph Design

We construct a graph from the data where nodes represent animals and edges denote familial relationships. An edge connects two nodes if the corresponding animals are part of the same family.





Accuracy in k-fold cross-validation



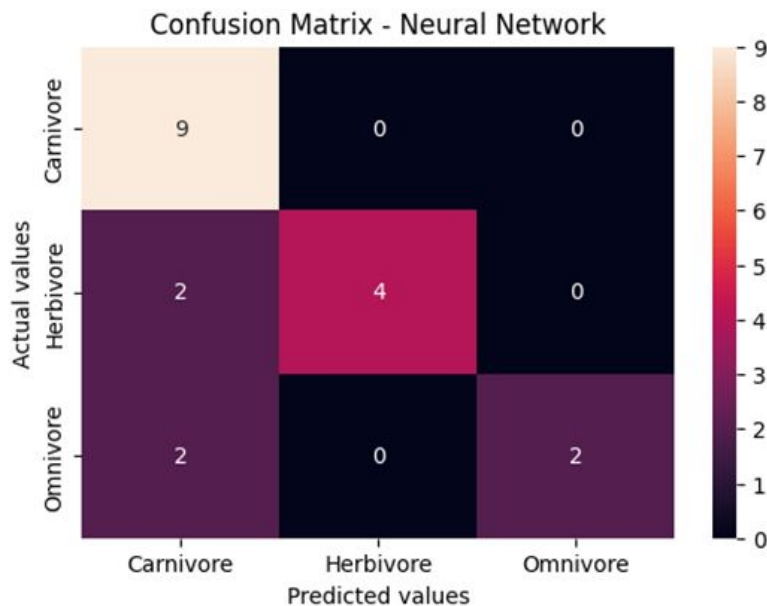


Accuracy in test data

	precision	recall	f1-score	support
Carnivore	0.69	1.00	0.82	9
Herbivore	1.00	0.67	0.80	6
Omnivore	1.00	0.50	0.67	4
accuracy			0.79	19
macro avg	0.90	0.72	0.76	19
weighted avg	0.85	0.79	0.78	19

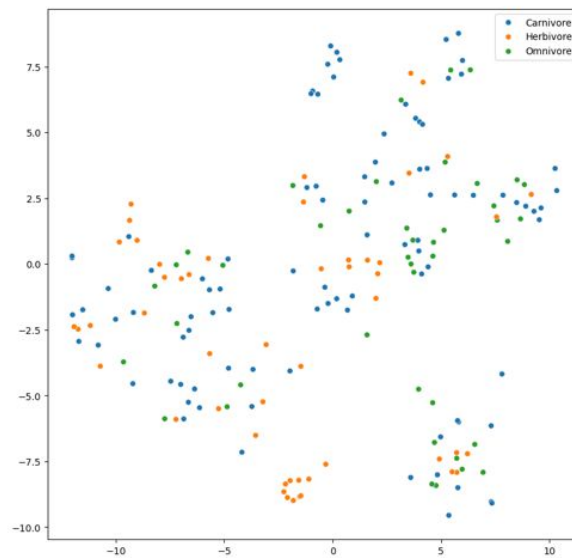
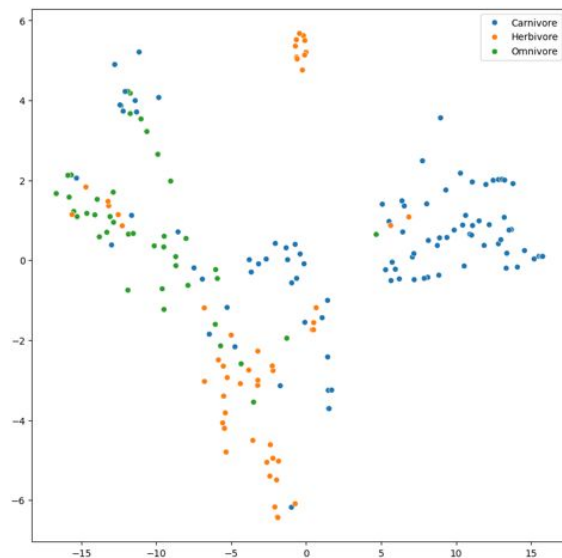


Confusion Matrix in test data



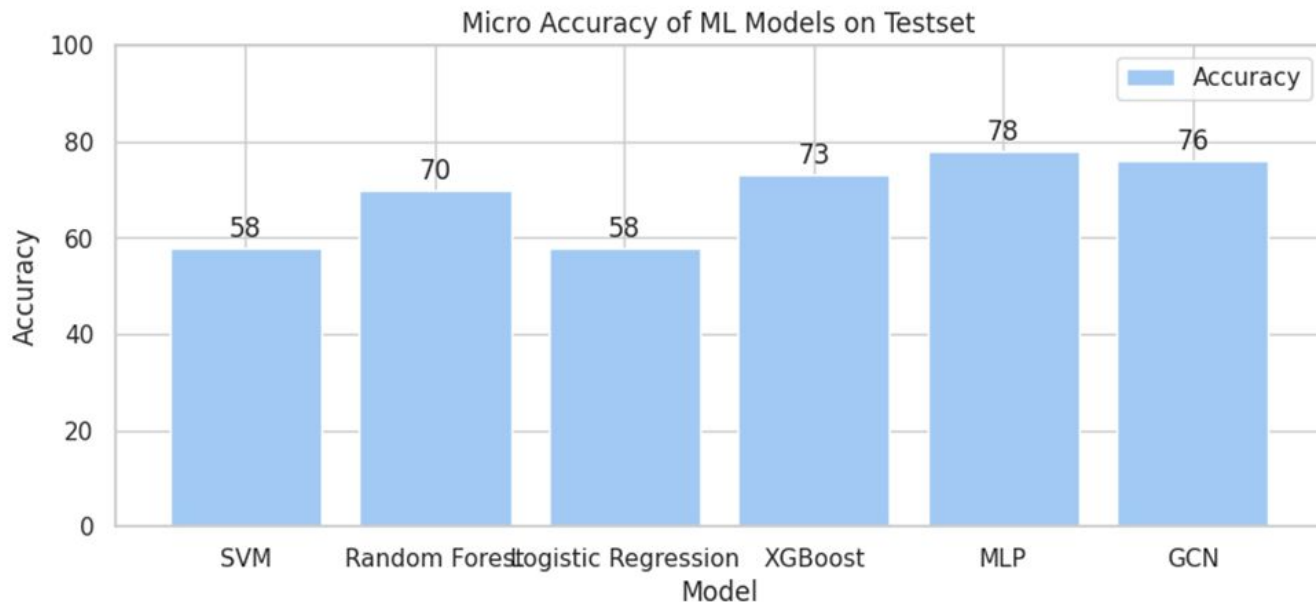


Visualizing node representations using t-SNE





Comparison of micro accuracy





Part 6: Conclusions

Key objectives:

- Summarize Key Points
- Highlight Implications



Conclusions

- We introduce a Graph Convolutional Network (GCN) model for classifying animal diets and compare its performance with other machine learning methods.
- The GCN model demonstrates high accuracy, indicating its effectiveness in classifying the dataset.
- Given the small sample size relative to the number of animal species, overfitting is a significant concern. To address this, we employ k-fold cross-validation and an external test set.



Thank You

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